

Accessible Delivery of Visual-Acoustic Biofeedback for Speech Sound Disorder

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Abstract

This paper presents staRt, an iOS and web application for visual-acoustic biofeedback aiming to facilitate widespread uptake of technology-enhanced speech training. staRt provides a real-time linear predictive coding (LPC) spectrum that can be compared to an adjustable slider representing a target frequency for vocal tract resonance. The software currently presents exercises for children with speech sound disorder who have difficulty pronouncing “r” sounds in English, but it shows promise for adaptation to a number of other contexts.

Index Terms: biofeedback, human-computer interaction, speech sound disorder

1. Introduction

New technologies have the potential to revolutionize the clinical management of childhood speech sound disorder (SSD), but uptake of new evidence-based practices has historically been limited by barriers of cost and the need for specialized training. This paper provides a technical description of staRt, an iOS and web application that aims to increase the accessibility and inclusivity of speech training incorporating visual-acoustic biofeedback. The software has been downloaded nearly 20,000 times by speech-language pathologists (SLPs) and families of children with SSD worldwide. We want to share the project with the Interspeech community at this time because we are re-envisioning the acoustic processing at the core of our tool and welcome feedback to guide our next steps.

2. Theoretical Background

2.1. Biofeedback for Speech Sound Disorder

In SSD, deficits in spoken communication negatively impact intelligibility and pose a barrier to participation across academic and social domains, with negative ramifications for educational and occupational outcomes [1]. Most children with SSD go on to develop typical speech by 8-9 years of age, but up to 5% of older children and adolescents continue to struggle with challenging sounds such as the American English “r” sound, /r/ [2].

Our team’s laboratory research has shown that children who have not responded to conventional speech therapy can benefit from treatment incorporating visual biofeedback [3]. We define biofeedback as the use of instrumentation to create a real-time display of speech that can be compared with a visual target representing the desired pronunciation. The visual display offers a new window of insight into a largely subconscious process and can also support learners who have difficulty hearing the difference between the target sound and their current production. Despite positive research findings, clinical uptake of biofeedback was limited for many years by the high cost of the technology

used in laboratory research. The staRt project was launched with the goal of expanding access to an evidence-based treatment method for children with speech difficulties. Originally developed for iOS, staRt is now accessible as a web application. Here, we focus on the web version that runs on standard web browsers, making the software device-agnostic and therefore accessible to a wider range of users. staRt requires no equipment besides a microphone and a device that can run a conventional web browser.

For vowels and vowel-like consonants like /r/, visual-acoustic biofeedback shows the resonant frequencies of the vocal tract (formants). In staRt, this information is provided in the form of a real-time LPC spectrum, with frequency on the x-axis and amplitude on the y-axis; formants appear as peaks in the spectrum. Figure 1 shows the acoustic signature of an adult-like /r/ (teal): a lowered frequency of the third formant (F3), which closely approximates the second formant (F2). This contrasts with the high F3 frequency seen in a distorted /r/ sound (blue). Figure 1 also shows the starfish slider used to set a target frequency for F3; it defaults to a value based on normative data from typical child speakers but can be manually adjusted.

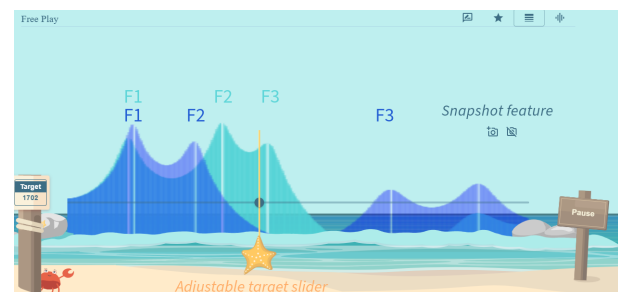


Figure 1: LPC spectrum showing the first three formants in the context of correct (teal) and distorted (blue) /r/ sounds. The blue wave was generated with the wave snapshot feature.

3. Technical Description

3.1. System architecture

The core of staRt is a real-time LPC spectrum implemented in JavaScript. At the present time, it is supported only in the Chrome browser. LPC is a source-filter model of the voice, where the glottis is modeled by noise (or pulse train) and the vocal tract is modeled by an infinite impulse response (IIR) filter. In the frequency domain, LPC can be summarized by the equation below, where $X(n)$ is the speech signal, $H(z)$ is the filter, and $E(z)$ is the noise source [4]:

$$X(z) = H(z)E(z) \quad \text{where:}$$

$$H(z) = \frac{1}{A(z)} = \frac{1}{1 - \sum_{k=1}^p a_k z^{-k}}$$

$H(z)$ estimates the spectral structure of the vocal tract via calculation of the LPC coefficients a_k , derived using Levinson-Durbin recursion on a given audio buffer [5].

The steps in audio processing in staRt are shown in Figure 2. In the web context, it was necessary to disable echo cancellation and automatic gain control settings that are typically active by default in the browser. Parameters including audio buffer size, sampling rate, bit resolution, and LPC order were chosen by the developers to optimize spectral envelope accuracy, CPU usage, and visualization smoothness. The animation uses a simplified peak-picking approach that detects local maxima exceeding a preset magnitude threshold (visible as the white lines superimposed over the spectrum in Figure 1). More sophisticated formant tracking can be done using numerical methods (Durand-Kerner) but only with lowered real-time potential.

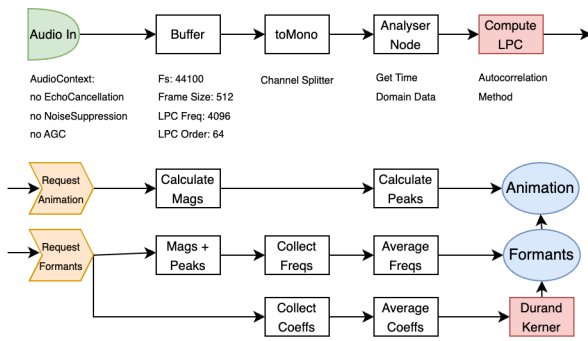


Figure 2: Audio signal flow of staRt app

3.2. User-facing features

staRt is intended for use by a child learner under the guidance of an SLP. A number of features aim to make the LPC visualization understandable and motivating in the clinical setting. All of the features listed below have been implemented in the iOS app and are in the process of being implemented in the web app.

1. An interactive tutorial explains how to interpret the LPC spectrum and provides examples of correct and incorrect /t/.
2. Clinicians score each production with a button or keypress.
3. “Word Quest” and “Syllable Quest” routines present a standard set of targets in randomized order. In “Word Quest”, stimuli are presented in an adaptive fashion based on clinician ratings, i.e. with task difficulty increasing when accuracy is high and decreasing when it is low.
4. Learners receive gamified reinforces, including coins added to a virtual stack after each trial and pop-ups for high scores.
5. Profiles stored in Firebase allow clinicians to track client progress across multiple devices.

3.3. staRt for telepractice

Although most SLPs use staRt while sitting side-by-side with a child client, a growing number of SLPs deliver their services via remote video calls (telepractice). We initially explored screen-sharing of the staRt web app over Zoom or similar platforms. However, this yielded a suboptimal user experience due to the latency and loss of frequency resolution introduced in audio streaming. Built-in noise suppression algorithms also pose a challenge when speech sounds are drawn out, as is commonly the case in speech treatment. To circumvent these limita-

tions, the staRt web interface has been integrated with WebRTC videoconferencing technology using the Vonage Video API. In the telepractice context, our software analyzes the speech signal on the client’s local device and transmits the data via web sockets, re-drawing the spectrum on the clinician’s screen with no loss of frequency resolution and minimal lag.

4. Future Work

We aim to expand staRt to reach other populations (e.g., learners of English as a second language) and speech targets (e.g., sibilant sounds like “s” and “sh”). We recently prototyped an alternative display in the form of a Discrete Fourier Transform (DFT) spectrum, which is more appropriate for the visualization of sounds with primarily aperiodic energy, like sibilants. With this addition, we can position staRt as a multi-purpose tool offering biofeedback for a range of challenging speech sounds.

Additionally, users have requested a simplified display that could be easily understood by younger children (the staRt app is recommended for ages 8 and up). They have also asked for automated feedback informing the user when they are approaching their target (e.g., by changing the color of the display) instead of relying on visual comparisons between the spectrum and the target. However, our current methods for peak-picking and formant-tracking are insufficiently stable to support these automated features across a range of vocal tract sizes. We are currently comparing alternative formant estimation models, including Kalman filtering [6] and neural networks [7, 8], with the goal of supporting more accurate formant tracking without introducing excessive compute times or processing costs.

5. Acknowledgments

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6. References

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